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EXEMPLARY BIFURCATION SEQUENCES

A bifurcation sequence is a sequential occurence of intersting bifurcation events in a family of maps, seen as the parameter is varied. In the next four chapters we describe four particularly exemplary sequences. Each is taken from the recent literature and illustrates a basic type of bifurcation behavior. Each chapter is devoted to one of these types, and may begin with an introduction to the basic concepts in I D and 2D. The four types of bifurcation are:

- absorbing areas,
- holes,
- fractal boundaries, and
- contact bifurcations.

C4 ABSORBING AREAS

We begin with a brief introduction to the concept of absorption in one and two dimensions, and then study an exemplary bifurcation sequence.

4.1 ABSORPTION CONCEPTS, I D

We introduced in Chapter 2 the notions of critical points, which bound zones of multiplicity, and trapping intervals, which are mapped into themselves. These notions come together in the concept of an absorbing interval. This is an interval in the domain of the map which is trapping, is bounded by critical points, and is super-attracting, which means that every point sufficiently close to the critical endpoints will jump into the absorbing interval after a finite number of applications of the map. In the context of an iterated map of an interval, the interesting dynamics take place within absorbing intervals. The critical points of a one-dimensional map determine absorbing intervals, and are useful in characterizing some bifurcations, especially those called global bifurcations. Examples of global bifurcations occur in Chapter 7.

4.2 ABSORPTION CONCEPTS, 2D

In two-dimensional iterations, we have a notion of absorbing area, generalizing the absorbing intervals of the one-dimensional case. The critical curves of a map of the plane playa role analogous to that of the critical points of a one-dimensional map: they are useful in determining absorbing areas. We will now illustrate the role of critical curves in determining these important areas in which the interesting dynamics occur. In this chapter we will study the first family of quadratic maps defined in the Introduction. We will now use the map of this family determined by b = - 0.8 to illustrate the use of critical curves to determine an absorbing area.

Our map has two fixed points, P and Q. The basic critical curve, L, is the locus of points with "coincident preimages," that is, the set of points having nearby points with different numbers of preimages or rank 1 (see Appendix 3, especially A3.2 - A3.3, for definitions). The fundamental critical curve L_{-1} is defined as the preimage of L. Thus, L is the image of L_{-1} under the map. Similarly, the derived critical curve LI is the image of L, and so on. All of these are called critical curves, as described in Chapter 2.

Note for those who have studied vector calculus: In the context of a generic smooth map, the fundamental critical curve L_{-1} will be a subset of the set of critical points in the Jacobian sense, points at which the Jacobian derivative of the map (a linear transformation) is degenerate (not a linear isomorphism), while the basic critical curve L is a subset of the set of critical values in the Jacobian sense. Inflection points are Jacobian critical points which do not belong to L_{-1}.

For this particular map, L_{-1} is the vertical axis, x = 0, and L is a horizontal line, y = b = - 0.8. It will be convenient to choose a bounded interval in L_{-1}, $S--1$ with endpoints a_{-1} which is the origin (0, 0), and ao' which is the image of a_{-1} under the map, the point (0, - 0.8).

Note: The critical curve denoted by L in the text is denoted by Lo in the figures. Figure 4-1 shows all these features and more. Note the points a_{-1} and a_0 and the segments of L$'_I L, LI, 00', L4AsthemapmovesL_I toL, andthepointa_I toao'LismovedtoLI'andthepoint$

tangency of LI and L2 at a2' and so on. Note that the curve L2 crosses L$_I atthepointPo.soL3istangenttoLatthepointPI'theimageofPo.andsoon.Similarly.thecurveL3crossesL_I$
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Our next goal is to use these critical curves to discover an absorbing area of the map. By definition (see Appendix A3.5), an absorbing area is a region of the plane such that: • it is mapped into (or onto) itself; • its boundary is made up of segments of critical curves, or of Ii,mit points of an infinite sequence of critical curves; and • it has a neighborhood every point of which eventually moves into the absorbing area. As an example, note in Figure 4-1 that the arcs b 1 a 1 of L, a 1 a2 of LI ' a2a3 of L2, a3a4 of L3 ' and a4bl of L4 bound a region d', shown shaded in Figure 4-1. This shaded region happens to be an absorbing area! First of all, it is invariant. For example, the segment a4bl of L4 is on the boundary of the region, d', but the image of this arc, aSb2 , is internal to d'. This is clearly shown in Figure 4-2, in which the image of a4bl is indicated in Ls . Also, d' is absorbing: A point external to d' is mapped into d' in a finite number of iterations, as shown in Figure 4-3, and this is the case for all points sufficiently near to d' .

Figure 4-2. also shows a shaded area, d , bounded by the critical curve segments introduced in Figure 4-1. It surrounds a hole, W, in which the fixed point Q is located. This is a smaller absorbing area, and is called an annular absorbing area for obvious reasons. Generally, absorbing areas may be topologically more complex, with many holes, and with many separate pieces. Absorbing

areas must contain attractors. One, d, is shown as a cloud of dots in Figure 4-3. It is the attracting set of d', and of the smaller absorbing area, d , as well. Usually there are smaller and smaller absorbing areas around any attractor of the map. All these, by definition, are bounded by critical arcs. And as they get smaller and smaller, but always enclose the same attractor, it is to be expected that the attractor itself is bounded by critical arcs, or by limit points of infinite sequences of critical arcs. Figure 4-4 shows 25 iterates of two intervals of $L'_l thetwopiecesofd('')L_l.TheseiteratesboundtheattractorshowninFigure4-$
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$HopfbifurcationoccursThefixedpointQbecomesarepellorandthecurverappearsasanattractiveinvarianto$

Stage 2: b = - 0.5 In Figure 4-6, we see the point repellor Q within the attractive cycle r , and surrounding that, our first absorbing area, d' , shown shaded in Figure 4-7. This area is mapped into itself, is bounded by arcs of critical curves, and is attractive (see Appendix 3.5 for the precise definition). In this case, the absorbing area is bounded by the arcs of the critical curves, L, Ll ' L 2 , L3 ' and L 4. These bounding arcs are generated by successive iterations of the map, as we now describe. Notice in Figure 4-7, which shows the critical arcs in more detail, that $L_l andLarestraightlines,crossingorthogonallyinonepoint.Letaodenotethispoint,(0,-0.5).Sinceitbelo$
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Stage 4: b = - 0.72 At this stage there is an attractive periodic cycle of period 11. The points of this cycle are labelled in iteration sequence in Figure 4-14. This II-cycle persists as b decreases further, through very many more bifurcations. The movie on the CD-ROM reveals an astonishing number of these, and many more have been observed, even in an interval of b values as narrow as 0.001. Stage 5: b= -0.78 At this stage we find two attractors coexisting within the annular absorbing area, a 28-cycle and an II-piece chaotic attractor. These are shown in Figure 4-15. Near b = 0.798, there is an explosion to a chaotic attractor filling an annular absorbing area.

Stage 6: b = - 0.7989995 Figure 4-16 shows the chaotic attractor, bounded by critical curves. Passing below b = - 0.8, there are a number of additional bifurcations which have been studied on the research frontier. Some of them will be described later in this book. Approaching b = - 1.0, further explosions are found. Stage 7: b = - 0.975 The densely dotted region of Figure 4-17 is a large chaotic attractor, an annular chaotic area, d. The frontier, F, of its basin of attraction, includes the inset of the 2-cycle, Ql ' Q2 . The boundary of d is very near F. This figure shows that the critical curves (on the boundary of the former chaotic area) are about to touch (and then to cross) the inset of the 2-cycle.

The enlargement of Figure 4-18 shows that, in addition, a contact of the frontier, F, with the boundary on L is about to occur. This contact bifurcation, described in Chapter 7, has the effect of destroying the chaotic attractor, or rather, of transforming it into a chaotic repellor. Now, almost all of the trajectories diverge to infinity, except for a Cantor set surviving inside the former chaotic area. A rough idea of the basin of infinity, D(00), is shown in Figure 4-19 as a black area. The basin of infinity includes infinitely many holes in the former absorbing area, d', only a few of which are shown in this figure. The light area is the basin of attraction of the attractor d. An enlargement is shown in Figure 4-20.